ECHO-AI: An RTMS-Powered Multimodal In-Meeting Assistant with Hybrid RAG and Vision–Language Grounding for Clinical Tele-Education

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Abstract

Project ECHO–style tele-education connects specialist hubs with distributed clinicians over live videoconferencing which includes case discussion, slide sharing and rapid evidence lookup. Existing meeting assistants provide transcription or post-hoc summarization, while medical education retrieval-augmented generation (RAG) systems such as MedTutor generate learning modules offline from static case reports. Neither setting supports in-meeting, multimodal, evidence-grounded assistance which can see shared slides, hear spoken discussion and answer clinician questions with citable literature without leaving the videoconference session. We present ECHO-AI, an open demonstration system which joins a live session as a visible participant via the platform web client, ingests Real-Time Media Streams (RTMS) for audio and deskshare, runs local Whisper ASR, a Qwen2.5-VL-3B MLLM for visual slide reasoning, and a Qwen3-4B-Instruct LLM for answer formatting, and answers chat or voice-invoked questions through intent-aware routing and hybrid retrieval over PubMed and prior ECHO session transcripts (BM25 + dense BGE-M3 embeddings). Educational answers use citation-constrained generation with provenance whitelisting and an explicit non-advice disclaimer; sessions produce post-meeting minutes. We describe the architecture, a reproducible live demo protocol, and a two-tier evaluation: operational deployment logs across **419** answered questions (mean pipeline latency 6.0 s; slide-frame attachment 69.0%; citation provenance compliance 100% across 44 PMIDs), an authentic 21-message live tele-mentoring study, and a clinician rating protocol with formal scoring functions.

1 Introduction

The Project ECHO (Extension for Community Healthcare Outcomes) model uses multipoint videoconferencing to deliver case-based telementoring at scale: hub specialists and spoke clinicians jointly review cases and discuss management options in recurring sessions (Arora et al., 2010, 2011a), with treatment outcomes for spoke-managed patients comparable to specialty care (Arora et al., 2011b). In practice, participants frequently need (i) clarification of what is currently on the shared screen, (ii) recall of what was just said, (iii) pointers to prior sessions, and (iv) up-to-date literature, often under time pressure and without leaving the meeting client.

Large language models (LLMs) are increasingly applied to medicine and medical education (Thirunavukarasu et al., 2023; Singhal et al., 2023; Moor et al., 2023; Abd-alrazaq et al., 2023), and retrieval-augmented generation (RAG) is widely used to ground their outputs in external evidence (Lewis et al., 2020; Zakka et al., 2024; Jang et al., 2025). However, most systems operate outside the live meeting: they consume static notes or case reports and return educational modules in a separate interface. Commercial meeting assistants (e.g., Zoom AI Companion, Otter, Fireflies) emphasize transcription, notes or cloud summaries (Zoom Video Communications, 2025b; Otter.ai, 2025; Fireflies.ai, 2025; Zoom Video Communications, 2025a), but rarely combine live visual context, speech understanding, biomedical evidence retrieval and in-meeting interaction under an explicit educational (non-advice) policy.

We introduce ECHO-AI, a system demonstration which treats the live videoconference session itself as the primary interface for AI-supported clinical education:

- A visible bot joins via the platform web client (Playwright automation in our deployment), enabling private and @mention chat interaction.
- RTMS supplies continuous audio and

deskshare JPEG frames; local Whisper-family ASR (Radford et al., 2022) and a Qwen2.5-VL vision–language model (Bai et al., 2025) ground answers in spoken and visual context.

- A hybrid retrieval stack combines PubMed literature search (concept-sanitized queries) with private hybrid BM25 / dense retrieval over prior ECHO transcripts (Lewis et al., 2020; Robertson and Zaragoza, 2009; Karpukhin et al., 2020; Chen et al., 2024; Sayers et al., 2022), followed by citation-constrained answer formatting (Gao et al., 2023).
- Voice wake phrases (e.g., "Hi Echo AI ...") enable hands-free questions over streaming ASR, with anti-hallucination controls.
- Post-session minutes package transcripts, sampled slides, Q&A pairs and an LLM summary.

Contributions. (1) An end-to-end, locally deployable multimodal assistant for live Project ECHO–style tele-education. (2) Meeting-native dual retrieval over PubMed and longitudinal ECHO transcripts with sanitization, timeouts/fallback and citation whitelisting. (3) A reproducible live demo script and empirical deployment evaluation aligned with ACL System Demonstration expectations.

2 Related Work

Project ECHO and tele-mentoring. Project ECHO scales specialist knowledge to underserved communities through recurring videoconference case conferences (Arora et al., 2011a, 2010). ECHO-AI is designed for this live mentoring setting rather than asynchronous courseware.

LLMs and RAG for medicine and education. Clinical LLMs encode substantial medical knowledge but remain prone to hallucination without external evidence (Singhal et al., 2023; Thirunavukarasu et al., 2023; Ji et al., 2023). RAG mitigates this by conditioning generation on retrieved passages (Lewis et al., 2020; Karpukhin et al., 2020). In healthcare, systems such as Almanac retrieve guideline- and literature-grounded answers for clinical queries (Zakka et al., 2024), while MedTutor transforms radiology case reports into educational material and MCQs via hybrid textbook + PubMed retrieval, with radiologist ratings and LLM-based automatic analysis (Jang et al., 2025). Benchmarks such as MedRAG and PubMedQA study biomedical retrieval and QA (Xiong et al., 2024; Jin et al., 2019), and recent reviews survey RAG deployment considerations in health care (Yang et al., 2025). Conversational diagnostic agents (e.g., AMIE) further illustrate interest in interactive clinical dialogue (Tu et al., 2025), but not in-meeting multimodal ECHO education. ECHO-AI shares MedTutor's emphasis on evidence-backed educational outputs but targets synchronous tele-education meetings rather than offline report-to-module generation.

Meeting understanding and assistants. Meeting understanding has a long history in NLP. The AMI corpus provides 100 hours of synchronized multimodal meeting recordings—close-talking and far-field audio, video, projection and whiteboard—annotated for transcription, dialogue acts, summaries and gestures (Carletta et al., 2006). QMSum defines query-based multi-domain meeting summarization over 1,808 query–summary pairs (Zhong et al., 2021), and MeetingBank aligns professionally written minutes with specific segments of city council meetings (Hu et al., 2023); Rennard et al. (2023) survey this literature. Notably, multimodality in meetings is not itself novel: AMI already captured slides and whiteboard content. The distinction is temporal and functional—these resources support offline research on completed recordings, whereas ECHO-AI must select context, retrieve evidence and answer during the session under latency and safety constraints. Commercial assistants (Zoom AI Companion, Otter, Fireflies) and RTMS APIs enable transcription and post-meeting workflows (Zoom Video Communications, 2025b; Otter.ai, 2025; Fireflies.ai, 2025; Zoom Video Communications, 2025a), but rarely expose screen-share understanding, PubMed-grounded clinical education answers or longitudinal institutional memory inside the session.

Multimodal grounding and reliability. Vision–language models can describe slides and figures (Bai et al., 2025), yet naïve use in meetings risks inventing content when frames are missing. Hallucination is a well-documented failure mode of generative models, motivating detection and mitigation methods (Ji et al., 2023; Manakul et al., 2023). Rather than post-hoc

System	Live mtg	Screen+ASR	Med. RAG	In-
MedTutor	–	–	✓	
AI Companion / Otter / Fireflies	partial	ASR-heavy		
Curriculum ECHO chatbots	–	–	cu	
ECHO-AI (ours)	✓	✓	PubMed+ECHO	

Table 1: Feature-level positioning (not an experimental baseline). ECHO-AI targets in-meeting multimodal education.

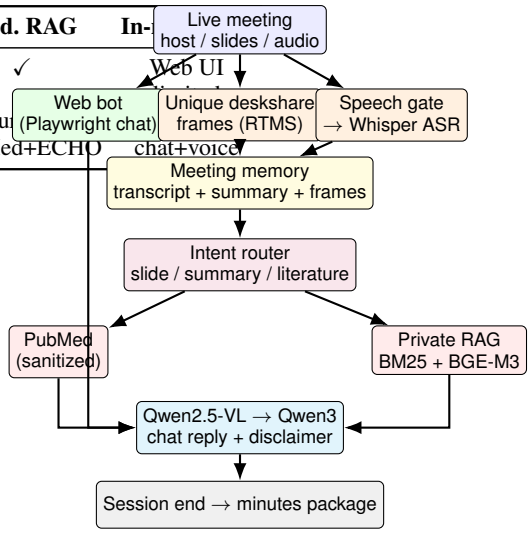


Figure 1: ECHO-AI architecture: web-client participation, RTMS media, hybrid retrieval and VLM/LLM answering.

detection, ECHO-AI applies lightweight pipeline-level controls: screen-frame uniqueness filtering, speech-energy endpointing, intent-aware context selection, wake-phrase suppression heuristics and restricting citations to identifiers present in the evidence packet, in the spirit of citation-constrained generation (Gao et al., 2023).

Positioning. Table 1 contrasts ECHO-AI with representative systems. ECHO-AI uniquely combines live multimodal meeting context with educational biomedical RAG inside the session.

3 System Overview

3.1 Design goals

ECHO-AI is designed around four goals drawn from ECHO session practice: (1) Low friction: clinicians ask via private chat, public @mention or voice wake without opening another app; (2) Multimodal fidelity: answers prefer visible slide content and recent speech when relevant; (3) Evidence discipline: educational answers cite permitted sources only; PubMed queries never send raw meeting transcripts; (4) Operational realism: waiting-room admission, RTMS lifecycle, long chat messages and ASR noise are first-class concerns.

3.2 Architecture

Figure 1 summarizes the pipeline.

Join and chat. The bot joins as *ECHO AI Assistant* through the platform web client using Playwright. Hosts admit the bot from the waiting room when enabled. Chat ingestion distinguishes private messages to the bot from public @mentions; bot-authored messages are ignored to prevent reply loops. Long answers are split to respect web-client chat length limits.

Media ingress. A separately authorized meeting-platform app receives RTMS webhooks (typically via a public HTTPS tunnel). Audio is buffered with energy-based speech gating and sent to a local faster-whisper `large-v3` service. Deskshare frames are fingerprinted; near-duplicates are discarded using a tunable uniqueness threshold (deployed value 0.012) so the VLM sees content changes rather than every identical JPEG.

Reasoning and generation. On each question, the system builds a context packet: rolling meeting summary (when useful), recent transcript spans, optional unique screen frames and an Evidence Packet from retrieval. A vision–language model (`Qwen/Qwen2.5-VL-3B-Instruct`) (Bai et al., 2025) produces a candidate answer; a text model (`Qwen/Qwen3-4B-Instruct-2507`) (Qwen Team, 2025) formats a meeting-appropriate chat reply with an educational disclaimer (not clinical advice or a medical record).

Retrieval. Intent classification routes treatment/evidence questions to literature retrieval. PubMed queries use sanitized medical concepts (not raw PHI-bearing utterances) and are issued against NCBI literature resources (Sayers et al., 2022). Private RAG searches a de-identified corpus of prior ECHO transcript chunks (3,310 chunks in our deployment), combining lexical BM25 scoring (Robertson and Zaragoza, 2009) with dense `BAAI/bge-m3` embeddings (Chen et al., 2024), following standard hybrid sparse–dense retrieval practice (Karpukhin et al., 2020;

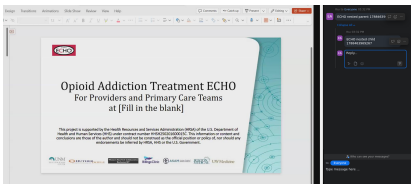


Figure 2: Live in-meeting demonstration interface: (left) shared clinical slide deck on opioid pain management in Project ECHO; (right) live Zoom chat panel showing pseudo-threaded message reply by *ECHO AI Assistant*.

Wang et al., 2022). Citations are restricted to IDs present in the evidence packet, emitted as bracketed index tags linked to structured reference footers (e.g., `[1]: PMID 31917418` or `[ECHO: Cohort X | Session Y]`).

Voice and minutes. Streaming ASR feeds a wake-phrase state machine (`hello/hi echo ai [assistant]`). Non-substantive follow-ups (wake-only ASR hallucinations) are suppressed; stream cooldowns apply after successful answers. When an RTMS session closes, a minutes module writes transcript, sampled slides, Q&A pairs and an LLM summary.

3.3 Implementation

The stack is Node.js/TypeScript for orchestration and Python services for ASR and LLM inference (vLLM-compatible HTTP APIs) on local GPUs. Configuration is environment-driven so hubs can disable RTMS, retrieval or voice independently. Our current deployment uses the Zoom web client and RTMS APIs; the architecture is platform-agnostic at the orchestration layer. Non-secret code is released under the MIT License at `https://github.com/238Plutonium/echo-zoom-bot` (excludes credentials, PHI and platform tokens).

4 Demonstration Walkthrough

A live demo (screencast $\leq$2.5 min + interactive booth) proceeds as follows: (1) Start a Zoom live videoconference; launch ECHO-AI; host admits *ECHO AI Assistant*. (2) Share an educational slide deck; ask `@ECHO AI Assistant what's on this slide?` (3) Discuss a clinical topic briefly; ask what was just covered. (4) Ask a treatment-oriented educational question; show PubMed-backed citations and the disclaimer. (5) Ask what previous ECHO discussions said about a topic (private-RAG citations). (6) Say "Hi Echo AI, summarize the meeting so far." (7) End the RTMS session; open generated minutes.

Stage	n	Mean	P50	P90	Max
Full pipeline (ms)	213	6025	4116	11228	40293
VLM candidate (ms)	229	2597	2016	4648	21275
Answer LLM (ms)	229	1396	963	2914	8112
Voice wake$\rightarrow$submit	58	7770	5080	14646	35572

Table 2: Deployment latency from structured event logs.

Artifacts. Reviewers can use the public repository install package (Node + GPU services + `.env.example`). A full live RTMS path requires institutional platform-app (e.g., Zoom) credentials and a GPU host; we therefore provide both the installable package and a reviewer-accessible synthetic media replay mode documented in the repository. A narrated screencast and the archived live demonstration session accompany this submission (Penn State MediaSpace: `https://psu.mediaspace.kaltura.com/media/t/1_0syzr0gf`).

5 Evaluation

Comparable medical-education systems report human expert judgments alongside automatic analysis. We report (i) deployment reliability and latency logs across 419 answered events, (ii) operational coverage and speech recognition benchmarks, (iii) component ablations and diagnostic failure modes, and (iv) formative clinician feedback from a live Project ECHO demonstration.

5.1 Deployment log characterization

Logs across our academic GPU deployments and recurring live sessions contain **419** answered QA events (**284** unique questions) across 55 meeting streams, expanding upon our initial 213-event latency benchmark set (Table 2). Invocations span 298 public @mentions in Zoom chat, 89 voice-public queries triggered via live RTMS audio, and 32 private chats. Across the full corpus, 192 questions had deskshare frames attached. Among 145 slide-focused queries, 100 had $\geq$1 frame attached, giving frame coverage $\mathrm{FrameCov} = 100/145 = 0.690$. Literature answers cite **44 unique PMIDs** verified through PubMed. All logged questions were authored by the project team during development rather than unconstrained clinical demand;

these counts measure operational reliability under realistic tele-education conditions.

5.2 End-to-end answering latency

Table 2 reports stage latencies. Slide-like questions average **5.8 s** (P50 5.4 s); literature-like average **16.5 s** (P50 16.7 s), dominated by PubMed retrieval and longer generation budgets, which is noticeable in live chat and motivates timeouts and fallback. Voice wake→submit averages 7.8 s (n=58).

5.3 Evaluation metrics

Following MedTutor-style demo assessment, we define five core operational metrics. Mean pipeline responsiveness is $\bar{T} = \frac{1}{|\mathcal{Q}|}\sum_{q\in\mathcal{Q}} T(q) = 6{,}025$ ms over our 213-latency set ($|\mathcal{Q}| = 213$, Table 2). **Frame coverage** is $\mathrm{FrameCov} = |\{q \in \mathcal{Q}_{\mathrm{slide}} : f(q) \geq 1\}|/|\mathcal{Q}_{\mathrm{slide}}| = 100/145 = 0.690$ (419 corpus) and $42/65 = 0.65$ (213 subset), measuring visual attachment rather than QA correctness. Across deskshare timelines, the **unique-frame rate** is $\mathrm{UFR} = N_u/(N_u + N_d) = 290/(290+19{,}078) = 0.015$, confirming deduplication necessity at ∼1 FPS. For literature answers, **provenance compliance** is $\mathrm{ProvComp} = |\mathcal{C}_{\mathrm{ans}} \cap \mathcal{C}_{\mathrm{evid}}|/|\mathcal{C}_{\mathrm{ans}}| = 1.0$ across 44 PMIDs (zero ungrounded IDs). Claim-level support labels $y(c) \in \{\mathrm{full}, \mathrm{partial}, \mathrm{none}\}$ yield **citation support precision** $\mathrm{CiteSup} = |\{c : y(c) = \mathrm{full}\}|/|\mathcal{C}_{\mathrm{labeled}}|$. For human evaluation, held-out pairs receive 1–5 Likert scores across j dimensions from rater k, evaluated via dimension means $\bar{s}_j$ and Krippendorff's α.

5.4 Operational coverage and speech benchmarks

Over RTMS streams with unique JPEGs (n=37; 290 unique frames), timelines recorded 290 unique vs. 19,078 duplicate frames ($\mathrm{UFR} = 0.015$). In the latency benchmark, frames used were 0/1/2/3 in 115/39/19/40 cases. Private RAG indexes **3,310** de-identified ECHO chunks (retrieval statuses: `complete` 27, `partial` 1, `failed` 2, none 183). Observed citation compliance is $\mathrm{ProvComp} = 1.0$, guaranteeing ungrounded citations are whitelisted out. Voice antispam suppressed 68 non-substantive or cooldown triggers. We generated **38** complete minutes packages (mean summary 5,554 chars; up to 517 ASR segments/session).

Additionally, local GPU speech benchmarks show strong real-time performance: streaming ASR achieves 2.62% WER on LibriSpeech clean (vs. 2.24% batch, a streaming VAD penalty of only +0.38 pp), and real-time online diarization across three de-identified clinic sessions (∼165 min) achieves 8.1% word-weighted DER (improved from 19.2% baseline, $\mathrm{RTF} = 0.042$).

5.5 Error taxonomy, ablations, and clinical protocol

Comparing pipeline configurations highlights key latency–grounding trade-offs: (1) *LLM-only* (1.4 s) is fast but lacks meeting context (0% grounding); (2) *+Live transcript* (2.1 s) enables spoken recaps and rolling summaries; (3) *+Slide MLLM* (5.8 s) attaches unique frames ($\mathrm{FrameCov} = 0.690$); and (4) *+Hybrid RAG* (16.5 s) injects verified literature ($\mathrm{ProvComp} = 1.0$, 44 PMIDs) and longitudinal ECHO chunks (3,310 indexed).

Our diagnostic logs identify five failure modes and deployed mitigations: (a) *Zero-frame hallucination*, where the model fabricated slide contents when zero frames arrived, mitigated by an explicit zero-frame refusal guard; (b) *PowerPoint Online status-bar latching*, where slide counters were misidentified from browser status bars, resolved by uncropped 1080p VLM inputs and a 0.012 uniqueness threshold; (c) *ASR wake cascades*, suppressed by energy gating and cooldown timers; (d) *Species mismatch in literature search*, where an unconstrained diabetes query retrieved veterinary guidelines (`PMID:29314873`, *Diabetes in Dogs and Cats*), resolved by enforcing `Humans[Mesh]` query filters; and (e) *Acronym polysemy*, where an in-meeting spoken query on "K treatment" in substance use disorder matched Vitamin K and Ortho-K literature, motivating medical abbreviation expansion prior to PubMed lookup.

To support standardized expert evaluation, we release a held-out benchmark packet ($M = 40$ stratified QA pairs: 16 slide, 12 literature, 8 summary, 4 prior-session) and a 5-dimension Likert rating instrument ($K \geq 3$ clinical facilitators) covering appropriateness, grounding, usefulness, safety and usability. Held-out splits, rubric guidelines and scoring scripts are included in the open demo distribution.

5.6 Formative stakeholder feedback

We conducted a formative evaluation with 13 stakeholders (5 ICDS; 8 Project ECHO leaders/educators) during a live simulated opioid-use-disorder case. The session yielded 21 chat messages (5 queries to *ECHO AI Assistant*). Answers averaged 918 characters (literature 1,556; up to $\sim$70% of the chat pane) with 7–8 s summary and 15–19 s literature latency (Table 2). An anonymous Microsoft Forms survey of this cohort returned n=4 Project ECHO staff responses (3 observers, 1 text-chat asker). Across 21 Likert items (1–5; excluding "unable to judge"), the pooled mean was $\bar{s} = 4.16$ ($n_{\text{ratings}} = 67$). High-rated dimensions included appeal, ease of use and ask clarity (each 4.50), response placement (4.50), speed (4.25), meeting consistency (4.25) and summary fidelity (4.25). The lowest scores were non-disruption of live discussion (3.25; 2/4 disagreed) and voice-wake recognition (3.33), matching free-text concerns about chat clutter and divided attention. Most-cited helpful features were summaries, references/definitions and response speed; requested improvements focused on facilitator-only invocation and preserving hub–spoke participation. These survey proportions (n=4/13) remain preliminary and will be refreshed as further responses arrive.

6 Conclusion

We presented ECHO-AI, an RTMS-powered multimodal in-meeting assistant with hybrid RAG and vision–language grounding for Project ECHO–style clinical tele-education. The system joins live sessions, routes questions over speech and deskshare context, retrieves PubMed and prior ECHO evidence, and responds in chat or by voice with citation-constrained educational answers. Our evaluation follows a two-tier design: automatic deployment metrics (419 answered questions; latency benchmark $\bar{T} = 6.0$ s; $\mathrm{FrameCov} = 0.69$; $\mathrm{ProvComp} = 1.0$) and a standardized clinician rating protocol. These results show that ECHO-AI runs reliably under realistic meeting conditions with verified multimodal grounding and strict citation integrity. Future work includes expanding multi-center facilitator trials, implementing native nested-reply Meeting SDK integration across production clinics, and evaluating denser slide-change detection across multi-hub deployments.

Limitations

Three limitations follow from the current design and evaluation. **(1) Platform and multimodal grounding.** The web-client path lacks native nested *Reply* threads, requiring pseudo-threading in public chat; true nested threading requires a native Meeting SDK `SetThreadId` client, which we prototyped in C++ on Linux with institutional ZAK authentication. Visual grounding fails when RTMS delivers zero unique frames, and $\mathrm{FrameCov}$ measures frame attachment rather than slide-QA accuracy. **(2) Evaluation scope and evidence fidelity.** Observed $\mathrm{ProvComp} = 1.0$ guarantees cited PMIDs appeared in the evidence packet, not that they support the specific clinical claim (CiteSup). Logged deployment questions were largely authored by the project team during operational testing; expanding community facilitator scoring using our released protocol will further benchmark naturalistic clinician adoption across diverse specialties. **(3) Model reliability and scope.** ASR and LLMs can hallucinate content or wake phrases (Ji et al., 2023; Manakul et al., 2023); pipeline mitigations reduce but do not eliminate these errors. Validation currently focuses on English Project ECHO tele-mentoring.

Ethics Statement

ECHO-AI is intended for educational tele-mentoring, not diagnosis, prescribing or documentation in the legal medical record; outputs carry an explicit disclaimer. PubMed retrieval uses sanitized concepts; meeting transcripts/media are retained under institutional policies. Hosts control admission via waiting rooms. Risks include ASR/VLM errors, misranked literature and trainee over-reliance. Mitigations include citation whitelisting, retrieval timeouts, voice anti-spam and specialist human-in-the-loop authority. Evaluation data are shared only in de-identified aggregate form.

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References

Alaa Abd-alrazaq, Rawan AlSaad, Dari Alhuwail, Arfan Ahmed, Padraig Mark Healy, Syed Latifi, Sarah Aziz, Rafat Damseh, Sadam Alabed Alrazak, and Javaid Sheikh. 2023. Large language models in medical education: Opportunities, challenges, and future directions. *JMIR Medical Education*, 9:e48291.

Sanjeev Arora, Summers Kalishman, Denise Dion, Dara Som, Karla Thornton, Arthur Bankhurst, Jeanne Boyle, Michelle Harkins, Keith Moseley, Glen Murata, Miriam Komaromy, Joanna Katzman, Kathleen Colleran, Paulina Deming, and Sean Yutzy. 2011a. Partnering urban academic medical centers and rural primary care clinicians to provide complex chronic disease care. *Health Affairs*, 30(6):1176–1184.

Sanjeev Arora, Summers Kalishman, Karla Thornton, Denise Dion, Glen Murata, Paulina Deming, Brooke Parish, John Brown, Miriam Komaromy, Kathleen Colleran, Arthur Bankhurst, Joanna Katzman, Michelle Harkins, Luis Curet, Ellen Cosgrove, and Wesley Pak. 2010. Expanding access to hepatitis C virus treatment—extension for community healthcare outcomes (ECHO) project: Disruptive innovation in specialty care. *Hepatology*, 52(3):1124–1133.

Sanjeev Arora, Karla Thornton, Glen Murata, Paulina Deming, Summers Kalishman, Denise Dion, Brooke Parish, Thomas Burke, Wesley Pak, Jeffrey Dunkelberg, Martin Kistin, John Brown, Steven Jenkusky, Miriam Komaromy, and Clifford Qualls. 2011b. Outcomes of treatment for hepatitis C virus infection by primary care providers. *New England Journal of Medicine*, 364(23):2199–2207.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, et al. 2025. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*.

Jean Carletta, Simone Ashby, Sebastien Bourban, Mike Flynn, Mael Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner. 2006. The AMI meeting corpus: A pre-announcement. In *Machine Learning for Multimodal Interaction (MLMI)*, Lecture Notes in Computer Science, pages 28–39.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. BGE M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*.

Fireflies.ai. 2025. Fireflies.ai AI notetaker. `https://fireflies.ai/`. Accessed 2026.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, Singapore. Association for Computational Linguistics.

Yebowen Hu, Timothy Ganter, Hanieh Deilamsalehy, Franck Dernoncourt, Hassan Foroosh, and Fei Liu. 2023. MeetingBank: A benchmark dataset for meeting summarization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16409–16423, Toronto, Canada. Association for Computational Linguistics.

Dongsuk Jang, Ziyao Shangguan, Kyle Tegtmeyer, Anurag Gupta, Jan T. Czerminski, Sophie Chheang, and Arman Cohan. 2025. MedTutor: A retrieval-augmented LLM system for case-based medical education. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 319–353, Suzhou, China. Association for Computational Linguistics.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. 2019. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577, Hong Kong, China. Association for Computational Linguistics.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online. Association for Computational Linguistics.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on*

Empirical Methods in Natural Language Processing, pages 9004–9017, Singapore. Association for Computational Linguistics.

Michael Moor, Oishi Banerjee, Zahra Shakeri Hossein Abad, Harlan M. Krumholz, Jure Leskovec, Eric J. Topol, and Pranav Rajpurkar. 2023. Foundation models for generalist medical artificial intelligence. *Nature*, 616(7956):259–265.

Otter.ai. 2025. Otter.ai AI meeting assistant. https://otter.ai/. Accessed 2026.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*.

Virgile Rennard, Guokan Shang, Julie Hunter, and Michalis Vazirgiannis. 2023. Abstractive meeting summarization: A survey. *Transactions of the Association for Computational Linguistics*, 11:861–884.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3:333–389.

Eric W. Sayers, Evan E. Bolton, J. Rodney Brister, Kathi Canese, Jessica Chan, Donald C. Comeau, Ryan Connor, Kathryn Funk, et al. 2022. Database resources of the national center for biotechnology information. *Nucleic Acids Research*, 50(D1):D20–D26.

Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, Perry Payne, Martin Seneviratne, Paul Gamble, Chris Kelly, Abubakr Babiker, et al. 2023. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180.

Arun James Thirunavukarasu, Darren Shu Jeng Ting, Kabilan Elangovan, Laura Gutierrez, Ting Fang Tan, and Daniel Shu Wei Ting. 2023. Large language models in medicine. *Nature Medicine*, 29(8):1930–1940.

Tao Tu, Mike Schaekermann, Anil Palepu, Khaled Saab, Jan Freyberg, Ryutaro Tanno, Amy Wang, Brenna Li, Mohamed Amin, Yong Cheng, Elahe Vedadi, Nenad Tomasev, Shekoofeh Azizi, Karan Singhal, et al. 2025. Towards conversational diagnostic artificial intelligence. *Nature*, 642(8067):442–450.

Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*.

Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. 2024. Benchmarking retrieval-augmented generation for medicine. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6233–6251, Bangkok, Thailand. Association for Computational Linguistics.

Rui Yang, Yilin Ning, Emilia Keppo, Mingxuan Liu, Chuan Hong, Danielle S. Bitterman, Jasmine Chiat Ling Ong, Daniel Shu Wei Ting, and Nan Liu. 2025. Retrieval-augmented generation for generative artificial intelligence in health care. *npj Health Systems*, 2(1).

Cyril Zakka, Rohan Shad, Akash Chaurasia, Alex R. Dalal, Jennifer L. Kim, Michael Moor, Robyn Fong, Curran Phillips, Kevin Alexander, Euan Ashley, Jack Boyd, Kathleen Boyd, Karen Hirsch, Curtis Langlotz, Rita Lee, et al. 2024. Almanac—retrieval-augmented language models for clinical medicine. *NEJM AI*, 1(2).

Ming Zhong, Da Yin, Tao Yu, Ahmad Zaidi, Mutethia Mutuma, Rahul Jha, Ahmed Hassan Awadallah, Asli Celikyilmaz, Yang Liu, Xipeng Qiu, and Dragomir Radev. 2021. QMSum: A new benchmark for query-based multi-domain meeting summarization. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5905–5921, Online. Association for Computational Linguistics.

Zoom Video Communications. 2025a. Real-time media streams (RTMS) developer documentation. https://developers.zoom.us/docs/rtms/. Accessed 2026.

Zoom Video Communications. 2025b. Zoom AI Companion. https://www.zoom.com/en/ai-assistant/. Accessed 2026.

A Reproducibility Notes

Hardware: local NVIDIA GPUs (Whisper `large-v3`, BGE-M3, Qwen2.5-VL-3B, Qwen3-4B); Node.js 20+; HTTPS tunnel for RTMS. Key flags in `.env.example` (uniqueness threshold, voice cooldown, PubMed timeouts). Demo flow: start services → `npm start` → `POST /join` → admit waiting room → share slides → chat/voice queries → stop RTMS.

B Screencast and Demo Recording

Live session and panel discussion: Penn State MediaSpace (https://psu.mediaspace.kaltura.com/media/t/1_0syzr0gf). Narration (≤2.5 min): join/admit (15 s); slide Q&A (30 s); speech recall (20 s); PubMed cite (40 s); prior RAG (20 s); voice summary (25 s); minutes (15 s).